

Bi-Lipschitz Absorption of c_0 Without a Linear Copy of c_0

OpenAI

September 26, 2026

Abstract

We construct a separable real Banach space Z that contains no linear copy of c_0 , yet is bi-Lipschitz equivalent to $Z \oplus_\infty c_0$. The same space contains a bi-Lipschitz image of every separable metric space.

1 Introduction

All Banach spaces in this paper are real. A map $F : X \rightarrow Y$ is a *bi-Lipschitz embedding* if there are constants $0 < c \leq C < \infty$ such that

$$c\|x - x'\|_X \leq \|F(x) - F(x')\|_Y \leq C\|x - x'\|_X \quad (x, x' \in X).$$

It is a *bi-Lipschitz equivalence* when it is also onto Y . The distinction between these two notions is central here: the result constructs an onto map between whole Banach spaces, while its linear obstruction concerns a subspace.

The nonlinear geometry of Banach spaces asks how much linear structure is retained by maps that control distances but need not respect addition. Ribe proved that uniformly homeomorphic normed spaces have the same finite-dimensional linear structure up to one common distortion bound [12, Theorem 1]. Aharoni and Lindenstrauss constructed nonseparable Banach spaces that are bi-Lipschitz equivalent but not linearly isomorphic, and asked for separable examples [2, pp. 281–282]. Ribe later produced separable uniformly homeomorphic nonisomorphic spaces, and Aharoni and Lindenstrauss obtained uniformly convex examples that are uniformly homeomorphic but not Lipschitz equivalent [13, 3]. Johnson, Lindenstrauss and Schechtman developed both rigidity results and separable examples with different linear structures under the weaker relation of uniform homeomorphism [8, Theorem 5.8 and Section 8(3)]. These results explain why a separable bi-Lipschitz counterexample requires more than a uniform deformation.

The space $c_0 = c_0(\mathbb{N})$ consists of real scalar sequences tending to zero, with the supremum norm. Godefroy, Kalton and Lancien proved that a Banach space bi-Lipschitz equivalent to a linear subspace of c_0 is linearly isomorphic to a subspace of c_0 . They also proved that a Banach space bi-Lipschitz equivalent to c_0 is linearly isomorphic to c_0 [6, Theorems 2.1–2.2]. These theorems concern the whole domain of an equivalence. They do not say that a Banach space containing a bi-Lipschitz image of c_0 must contain a linear copy of c_0 . Kalton recorded the latter question as Problem 2, separately from the separable Lipschitz-isomorphism question in Problem 3 [10, pp. 21–22]. Hájek, Johanis and Schlumprecht still described the embedding question as apparently open in their January 2024 preprint (arXiv:2401.00831v1), later published in 2025 [7].

For Banach spaces X, Y , write $X \oplus_\infty Y$ for their product with norm $\|(x, y)\| = \max\{\|x\|, \|y\|\}$. The main result is the following whole-space form of a negative answer to the c_0 embedding question.

Theorem 1.1. *There is a separable real Banach space Z containing no closed linear subspace isomorphic to c_0 , and an onto bi-Lipschitz map*

$$F : Z \oplus_\infty c_0 \longrightarrow Z.$$

The map is automatically a bijection by its lower Lipschitz bound. The two spaces in Theorem 1.1 cannot be linearly isomorphic: the direct sum contains a linear copy of c_0 , whereas Z does not. Restricting F to that summand gives a bi-Lipschitz embedding of c_0 into this same Z . Aharoni proved that every separable metric space embeds bi-Lipschitzly into c_0 [1]; composition therefore makes this one space Z a bi-Lipschitz universal target for separable metric spaces.

There are related separable constructions with weaker control of small distances. Kalton constructed a separable Schur space, in which weakly convergent sequences converge in norm, uniformly homeomorphic to a space containing complemented c_0 [9, Theorem 4.6 and Propositions 5.1–5.2]. The Schur property excludes a linear copy of c_0 . Interleaving the two c_0 summands also gives uniform c_0 absorption by that same Schur space, with a nonlinear modulus of continuity. Sari constructed a coarse-Lipschitz embedding of c_0 into a separable dual Banach space, which contains no linear copy of c_0 [14, Introduction and Theorem 5]. Coarse-Lipschitz distance bounds allow additive errors. Theorem 1.1 obtains global two-sided Lipschitz bounds and an onto map.

An earlier real-separable counterexample to the Lipschitz-isomorphism question uses a different linear obstruction: one space contains $c_0(\ell_2)$, the space of ℓ_2 -valued null sequences, and its bi-Lipschitz equivalent partner contains no linear copy of that space [11, Theorem 1.1]. Its Hilbert-block and localized Lipschitz-free argument is independent of the construction here. The present construction excludes ordinary c_0 and yields absorption by the same space.

Proof overview

The first step is algebraic. Suppose E, U are Banach spaces and $P : E \oplus_\infty U \rightarrow U$ is the projection. We seek an onto bi-Lipschitz map $g : E \oplus_\infty U \rightarrow U$, a bounded linear map $Q : Z \rightarrow U$, and a Lipschitz map $K : E \oplus_\infty U \rightarrow Z$ such that

$$QK = g - P. \tag{1.1}$$

Section 2 proves that these data make $F(b, t) = b + K(t, Qb)$ an onto bi-Lipschitz map $Z \oplus_\infty E \rightarrow Z$. Thus the construction has two concrete tasks: realize (1.1) with $E = c_0$, and arrange that Z contains no linear c_0 . The identity lifts the correction $g - P$; it imposes no surjectivity or right-inverse requirement on Q . This distinction matters because a Lipschitz right inverse of a linear quotient onto a separable Banach space always yields a bounded linear right inverse [5, Corollary 3.2].

We first construct g . Let E and U be countable-coordinate copies of c_0 . Section 3 constructs the finite-coordinate weights and their localized correction functions. Split the coordinates of U into finite blocks arranged in rows. In each block choose a vector v and a functional w with $wv = 1$. A scalar entering from the preceding block replaces the scalar wx of the current input block, while the rest of that block is unchanged. Denote this frozen shift at state s by S_s . With all v, w fixed, the output readouts recover the input by an explicit linear inverse. The weights actually depend on the normalized input, so this frozen inverse does not by itself invert the nonlinear map. Section 4 controls the change of the weights to obtain global bi-Lipschitz bounds, then uses Brouwer’s theorem on finite rows. An extra scalar at the end of each row both balances the dimensions and forces the possible output spillover to vanish. Density and a closed-image argument then give onto-ness.

We next construct Z and lift the correction. Record each coordinate of $(S_s - P)s$ as a scalar function of s in the unit ball of $E \oplus_\infty U$. These functions depend on finitely many coordinates

and have localized supports. Section 5 enlarges them by two local operations, one making a function constant on a small cylinder and one retaining only the change. A strict scale budget keeps their supports localized, and a nonincreasing order of scales controls what happens after a small local change. Tail factors give a second way to separate supports.

Section 6 pairs these tests with mass-zero finite combinations of points. This uses the molecule viewpoint of Arens and Eells [4, Section 2] and the Lipschitz-free framework of Godefroy and Kalton [5, Sections 1–2]. Here the norm is built from lists of selected tests whose signed sums have a common Lipschitz bound. It combines their evaluation norms, with exponents approaching one, in an outer square sum. The square sum prevents a hypothetical c_0 basis from escaping through different test classes; the varying exponents then select one scalar test detecting each basis vector. We prove directly the completion and duality properties needed for this norm. The coordinate tests define Q , and radial extension of the point molecules defines K with (1.1).

Finally, a hypothetical linear c_0 basis in Z has uniformly bounded signed sums. Section 7 extracts one detecting test for each vector from a single class with a common operation budget. A test’s label records its original correction coordinate. Distinct labels can be separated either by tails or by nested coordinate cylinders. For a repeated label, an expansion leaves either disjoint tails or a first small local change. In the latter case, its support shrinks and its value becomes small enough that nested tests still have uniformly Lipschitz signed sums. The norm then forces the signed sums of the vectors to grow, a contradiction. Section 8 combines this obstruction with (1.1) and the onto shift.

2 The absorption criterion

The passage from a lifted correction to an absorption map is elementary and independent of the coordinate construction. We state it first to specify exactly what that construction must supply. Triangular changes of variables also appear in the quotient-section construction of Aharoni and Lindenstrauss [2, pp. 281–282] and in the free-space splitting of Godefroy and Kalton [5, Theorem 2.12]. The following argument lifts a correction term and supplies its own inverse formula.

Lemma 2.1 (Absorption from a lifted correction). *Let E, U, Z be real Banach spaces, let $D = E \oplus_\infty U$, and let $P(t, u) = u$. Suppose $Q : Z \rightarrow U$ is bounded linear, $g : D \rightarrow U$ is a bi-Lipschitz bijection, and $K : D \rightarrow Z$ is Lipschitz. Assume the exact identity*

$$QK(t, u) = g(t, u) - u \quad ((t, u) \in D). \quad (2.1)$$

Then

$$F : Z \oplus_\infty E \longrightarrow Z, \quad F(b, t) = b + K(t, Qb)$$

is a bi-Lipschitz bijection. If

$$q = \|Q\|, \quad L_K = \text{Lip}(K), \quad \|g(x) - g(x')\|_U \geq c_g \|x - x'\|_D$$

for some arbitrary lower constant $c_g > 0$, then

$$\text{Lip}(F) \leq 1 + L_K \max\{1, q\}, \quad \text{Lip}(F^{-1}) \leq \max\left\{1 + \frac{L_K q}{c_g}, \frac{q}{c_g}\right\}. \quad (2.2)$$

No surjectivity or right-inverse assumption is imposed on Q .

Proof. For $d \in Z$, define

$$(t_d, u_d) = g^{-1}(Qd), \quad G(d) = (d - K(t_d, u_d), t_d). \quad (2.3)$$

This is defined for every d because g is onto U . Identity (2.1) gives

$$QF(b, t) = Qb + QK(t, Qb) = g(t, Qb).$$

Thus g^{-1} reads (t, Qb) from $QF(b, t)$, and (2.3) gives $G(F(b, t)) = (b, t)$. Conversely, write $(t, u) = g^{-1}(Qd)$ and $b = d - K(t, u)$. Then

$$Qb = Qd - QK(t, u) = g(t, u) - (g(t, u) - u) = u,$$

so $F(G(d)) = d - K(t, u) + K(t, Qb) = d$. This proves bijectivity with inverse G .

The map $(b, t) \mapsto (t, Qb)$ has Lipschitz constant at most $\max\{1, q\}$, which proves the first estimate in (2.2). For $d, d' \in Z$, the lower bound for g gives

$$\|(t_d, u_d) - (t_{d'}, u_{d'})\|_D \leq \frac{q}{c_g} \|d - d'\|_Z.$$

The two coordinates in (2.3) therefore have Lipschitz constants at most $1 + L_K q / c_g$ and q / c_g , respectively. Their maximum is the second estimate in (2.2), and its reciprocal is a lower Lipschitz constant for F . $\square$

The remaining sections supply this criterion with $E = c_0$. In particular, the inverse calculation uses only $QK = g - P$. It never asks for an element b with prescribed Qb until the formula $b = d - K(g^{-1}(Qd))$ has already produced one.

3 Coordinate weights and localized corrections

The shift will replace one scalar in each coordinate block by a scalar from the preceding block. We first describe this operation, then choose weights whose dependence on the input is slow and whose correction coordinates have the support properties needed for the later tests.

For a countably infinite set A , let $c_0(A)$ have its supremum norm. Write $\mathbb{N} = \{1, 2, \dots\}$, index levels by $\lambda = (i, n) \in \mathbb{N}^2$, and put $J_\lambda = i + n$. For $j \geq 1$ set

$$d_j = 2^{-j}, \quad \mathcal{A}_j = ([-1, 1] \cap d_j \mathbb{Z})^j,$$

and let $\mathcal{A}_0$ consist of one point. The finite coordinate block at λ is

$$\Gamma_\lambda = \{(\lambda, j, \xi) : 0 \leq j \leq J_\lambda, \xi \in \mathcal{A}_j\}.$$

We use

$$E = c_0(\mathbb{N}), \quad U = c_0\left(\bigsqcup_{\lambda \in \mathbb{N}^2} \Gamma_\lambda\right), \quad D = E \oplus_\infty U,$$

and write $x = (t, (x^\lambda)_\lambda)$. Put $P(t, u) = u$. Fix an enumeration of the disjoint union of the coordinate sets of E and U . The map $\pi_j : D \rightarrow \mathbb{R}^j$ retains the first j coordinates in this enumeration; π_0 is the unique map into $\mathbb{R}^0$. Write $B = \{s \in D : \|s\|_D \leq 1\}$. For $n > 1$ let $\lambda^- = (i, n - 1)$ and set

$$r_{i,1}(x) = \max\{|t_i|, \|x^{i,1}\|_\infty\}, \quad r_{i,n}(x) = \max\{\|x^{i,n-1}\|_\infty, \|x^{i,n}\|_\infty\}.$$

Each r_λ is 1-Lipschitz. Every input coordinate occurs in at most two of these maxima. Thus, for every fixed $x \in D$,

$$r_\lambda(x) \longrightarrow 0 \quad \text{as } \lambda \text{ leaves finite sets.} \quad (3.1)$$

Indeed, infinitely many radii at least $\varepsilon > 0$ would require infinitely many input coordinates of modulus at least ε .

The scalar replacement uses a vector $v \in \mathbb{R}^{\Gamma_\lambda}$ and a row $w \in (\mathbb{R}^{\Gamma_\lambda})^*$ with $wv = 1$. For a block vector $z \in \mathbb{R}^{\Gamma_\lambda}$ and an incoming scalar $b \in \mathbb{R}$,

$$z = (z - v w z) + v(w z), \quad z + v(b - w z) = (z - v w z) + v b.$$

The first summand belongs to $\ker w$, so the operation preserves it and replaces the scalar readout $w z$ by b . Along a row of blocks, we will take b from the preceding block's readout, with t_i entering the first block. Section 4 constructs the inverse for any fixed collection of such pairs. The conditions in (W1) below combine this scalar identity with uniform bounds on the vectors and rows.

The pairs will depend on a state $s \in B$. Condition (W2) makes their variation small after multiplication by the local radius, which bounds the size of the input blocks involved in the replacement. This is the estimate that will preserve the frozen shift's distance bounds when the weights vary with the input.

The bands j in each block serve a different purpose: localization. For $j \geq 1$, a correction coordinate indexed by (λ, j, ξ) will be supported where the first j state coordinates are close to ξ . Whenever $j < J_\lambda$, it will also require a positive lower bound on $r_\lambda(s)$ depending only on j . The first restriction becomes sharper as j grows; the second gives a fixed radial threshold when j is fixed and the level varies. We will prove these two support bounds for the correction coordinates after constructing the weights.

Lemma 3.1 (Weights at one level). *Let $\eta = 10^{-3}$. There are continuous maps*

$$v^\lambda : B \rightarrow \mathbb{R}^{\Gamma_\lambda}, \quad w^\lambda : B \rightarrow (\mathbb{R}^{\Gamma_\lambda})^*$$

depending on finitely many coordinates of their argument, such that

$$\|v^\lambda(s)\|_\infty \leq 1, \quad \|w^\lambda(s)\|_1 \leq 2, \quad w^\lambda(s)v^\lambda(s) = 1, \quad (\text{W1})$$

and, for $s, s' \in B$,

$$\min\{r_\lambda(s), r_\lambda(s')\} \max\{\|v^\lambda(s) - v^\lambda(s')\|_\infty, \|w^\lambda(s) - w^\lambda(s')\|_1\} \leq \eta \|s - s'\|_D. \quad (\text{W2})$$

More precisely, the j th band of either weight is the product of a coefficient $\theta_j^{J_\lambda}(r_\lambda(s))$ and a vector or row depending only on $\pi_j s$. If $\theta_j^{J_\lambda}(r) > 0$, then

$$r \leq \rho_j \quad (j \geq 1), \quad r \geq \ell_j \quad (j < J_\lambda), \quad (3.2)$$

for positive constants ρ_j, ℓ_j defined in the proof.

Proof. Fix $j \geq 1$ and $s \in B$. For $\xi \in \mathcal{A}_j$ put

$$b_\xi^j(s) = \left(1 - \frac{\|\pi_j s - \xi\|_\infty}{d_j}\right)_+, \quad B_j(s) = \sum_{\xi \in \mathcal{A}_j} b_\xi^j(s).$$

The grid includes both endpoints in every coordinate. Every point of $[-1, 1]^j$ is within $d_j/2$ in supremum norm of a grid point, including points on the boundary of the cube. Thus $B_j(s) \geq 1/2$; in particular, the following row is defined everywhere on B :

$$w_\xi^j(s) = \frac{b_\xi^j(s)}{B_j(s)}, \quad v_\xi^j(s) = \min\left\{1, \left(2 - \frac{\|\pi_j s - \xi\|_\infty}{d_j}\right)_+\right\}.$$

The entries are nonnegative, $\|w^j(s)\|_1 = 1$ and $\|v^j(s)\|_\infty \leq 1$. If $w_\xi^j(s) > 0$, then $\|\pi_j s - \xi\|_\infty < d_j$ and $v_\xi^j(s) = 1$. Consequently $w^j(s)v^j(s) = 1$. At band $j = 0$, set both the sole coordinate of v^0 and the sole entry of w^0 equal to 1.

Here are explicit finite Lipschitz bounds. Put $N_j = |\mathcal{A}_j| = (2^{j+1} + 1)^j$. Since each b_ξ^j and v_ξ^j has Lipschitz constant at most d_j^{-1} ,

$$\|b^j(s) - b^j(s')\|_1 \leq N_j d_j^{-1} \|s - s'\|_D.$$

Normalizing a nonnegative vector whose ℓ_1 norm is at least $1/2$ changes distances by at most a factor 4 in ℓ_1 . Indeed,

$$\left\| \frac{b}{\|b\|_1} - \frac{b'}{\|b'\|_1} \right\|_1 \leq \frac{\|b - b'\|_1 + |\|b\|_1 - \|b'\|_1|}{\|b\|_1} \leq 4\|b - b'\|_1.$$

Thus $L_j = 4N_j/d_j$ bounds both $\text{Lip}(w^j)$ in ℓ_1 and $\text{Lip}(v^j)$ in the supremum norm.

Choose $H > 0$ with $\pi/H \leq \eta/2$ and inductively choose numbers $\rho_j > 0$ so that

$$0 < \rho_1 < 1, \quad \rho_{j+1} < e^{-H} \rho_j, \quad L_j \rho_j \leq \eta/4 \quad (j \geq 1). \quad (3.3)$$

Each constraint at stage j has a positive right-hand side, so this choice is possible. Put $\ell_j = e^{-H} \rho_{j+1}$ for $j \geq 0$. For an integer $J \geq 1$ define $\theta^J(r) = (\theta_0^J(r), \dots, \theta_J^J(r))$ on $[0, 1]$ as follows. For $r \geq \rho_1$ take $\theta^J(r) = e_0$. As r decreases across $[e^{-H} \rho_j, \rho_j]$, $1 \leq j \leq J$, rotate from e_{j-1} to e_j by setting

$$\theta_{j-1}^J(r) = \cos \alpha_j(r), \quad \theta_j^J(r) = \sin \alpha_j(r), \quad \alpha_j(r) = \frac{\pi}{2H} \log \frac{\rho_j}{r},$$

with all other entries zero. Between these transition intervals keep the most recently reached unit vector, and for $0 \leq r \leq e^{-H} \rho_J$ take $\theta^J(r) = e_J$. The strict inequality in (3.3) makes the intervals disjoint and ordered. At each transition endpoint the formula agrees with the adjacent constant value; this also proves continuity at $r = 0$. Every entry is nonnegative,

$$\sum_{j=0}^J \theta_j^J(r)^2 = 1, \quad \sum_{j=0}^J \theta_j^J(r) \leq 2. \quad (3.4)$$

For $0 < r \leq 1$, the path $u \mapsto \theta^J(e^u)$ is piecewise continuously differentiable and has ℓ_1 speed at most $(\pi/(2H))(|\sin \alpha| + |\cos \alpha|) \leq \pi/H$. It is therefore π/H -Lipschitz in the variable $\log r$. The schedule also gives (3.2). For $j = 0$, only the second assertion applies. For $j = J$, only the first assertion applies, and $\theta_J^J(0) = 1$ is allowed.

For $J = J_\lambda$ and $r = r_\lambda(s)$, concatenate the bands:

$$v^\lambda(s) = (\theta_j^J(r) v^j(s))_{j=0}^J, \quad w^\lambda(s) = (\theta_j^J(r) w^j(s))_{j=0}^J.$$

Every band depends on finitely many coordinates: $\pi_j s$ and $r_\lambda(s)$ involve finite coordinate sets. Continuity includes the cases $r = 0$ and transition endpoints. The bounds on the band weights and (3.4) give

$$\|v^\lambda(s)\|_\infty \leq 1, \quad \|w^\lambda(s)\|_1 = \sum_{j=0}^J \theta_j^J(r) \leq 2, \quad w^\lambda(s) v^\lambda(s) = \sum_{j=0}^J \theta_j^J(r)^2 = 1.$$

This proves (W1).

For (W2), put $r = r_\lambda(s)$, $r' = r_\lambda(s')$, $m = \min\{r, r'\}$ and $\delta = \|s - s'\|_D$. If $m = 0$ there is nothing to prove. Otherwise

$$m |\log r - \log r'| \leq |r - r'| \leq \delta.$$

Change the band coefficients first, using the band weights at s' . The logarithmic Lipschitz estimate and the bounds $\|v^j(s')\|_\infty, \|w^j(s')\|_1 \leq 1$ show that this part of either norm difference,

multiplied by m , is at most $(\pi/H)\delta \leq (\eta/2)\delta$. Then change the band weights with coefficients fixed at $\theta^J(r)$. Only positive j vary, and $\theta_j^J(r) > 0$ implies $m \leq r \leq \rho_j$. Hence this part of either norm difference, multiplied by m , is at most

$$\delta \sum_{j=1}^J \theta_j^J(r) m L_j \leq \frac{\eta\delta}{4} \sum_{j=1}^J \theta_j^J(r) \leq \frac{\eta\delta}{2}.$$

The sum of the two contributions proves (W2). $\square$

We record the support information in the form used by the lifting construction. For $s \in B$ define

$$a_{i,0}(s) = t_i, \quad a_{i,n}(s) = w^{i,n}(s) s^{i,n}, \quad f_{(\lambda,j,\xi)}(s) = \theta_j^{J_\lambda}(r_\lambda(s)) v_\xi^j(s) (a_{i,n-1}(s) - a_{i,n}(s)). \quad (3.5)$$

These functions will be the coordinates of the shift correction. We first prove their support and continuity properties directly from the weights.

Lemma 3.2 (Finite dependence and local support). *For every $\gamma = (\lambda, j, \xi)$, the function f_γ is continuous, vanishes at 0, and depends on a nonempty finite set I_γ of coordinates of D . The sets can be chosen to include the first j coordinates and every coordinate used by r_λ . Uniformly in γ ,*

$$|f_\gamma(s)| \leq 4r_\lambda(s) \leq 4, \quad \text{Lip}(f_\gamma) \leq 4 + 12\eta. \quad (3.6)$$

Moreover

$$f_\gamma(s) \neq 0 \implies \begin{cases} \|\pi_j s - \xi\|_\infty \leq 2d_j, & j \geq 1, \\ r_\lambda(s) \geq \ell_j, & j < J_\lambda, \end{cases} \quad (3.7)$$

where both conditions are imposed when both apply.

Proof. For $n > 1$, $a_{i,n-1}$ uses only $s^{i,n-1}$, the local radius $r_{i,n-1}(s)$, and $\pi_{J_{i,n-1}} s$. The former radius uses $s^{i,n-2}$ when $n > 2$ and t_i when $n = 2$. The current scalar $a_{i,n}$ uses only $s^{i,n}$, $r_{i,n}(s)$, and $\pi_{J_{i,n}} s$; for $n = 1$ the predecessor scalar is t_i . Since $J_{i,n-1} < J_{i,n}$, one possible common finite set for all coordinates γ at $\lambda = (i, n)$ is the union of the first J_λ enumerated coordinates, the coordinate t_i , and every coordinate of the blocks $\Gamma_{i,k}$ with $k \in \{n, n-1, n-2\} \cap \mathbb{N}$. Call it I_γ . It is nonempty and contains the coordinates required in the statement.

The row bound (W1) gives $|a_{i,n-1}(s) - a_{i,n}(s)| \leq 4r_\lambda(s)$, also when $n = 1$. Thus $|f_\gamma(s)| \leq 4r_\lambda(s) \leq 4$ and $f_\gamma(0) = 0$. To verify the uniform Lipschitz bound, fix $s, s' \in B$ and write $\delta = \|s - s'\|_D$. At each level choose the pair $(\bar{v}^\lambda, \bar{w}^\lambda)$ from whichever of s, s' has the larger local radius. If $e \in \{s, s'\}$ is the other endpoint, (W2) gives

$$r_\lambda(e) \max\{\|v^\lambda(e) - \bar{v}^\lambda\|_\infty, \|w^\lambda(e) - \bar{w}^\lambda\|_1\} \leq \eta\delta;$$

the same bound is trivial if e supplied the frozen pair. For $e = s, s'$, replacing its weights by the frozen weights in the correction vector $v^\lambda(e)(a_{i,n-1}(e) - a_{i,n}(e))$ costs at most $4\eta\delta$ from the vector and at most $2\eta\delta$ from its current and predecessor rows together. At $n = 1$ the predecessor t_i does not change. With all weights frozen, the correction vector is linear in the input and has norm at most $4r_\lambda(s - s') \leq 4\delta$. Comparing the two endpoints therefore gives $|f_\gamma(s) - f_\gamma(s')| \leq (4 + 12\eta)\delta$.

If $f_\gamma(s) \neq 0$, its displayed product has both $\theta_j^{J_\lambda}(r_\lambda(s)) > 0$ and $v_\xi^j(s) > 0$. For $j \geq 1$, the formula for v_ξ^j gives $\|\pi_j s - \xi\|_\infty < 2d_j$, which implies the first closed inequality in (3.7). If $j < J_\lambda$, the second inequality follows from (3.2). This includes $j = 0$; no grid condition is imposed there. $\square$

4 The state-dependent shift is onto

The weights of Lemma 3.1 now provide the map required by the absorption criterion. We first invert arbitrary frozen weights, then bound their variation, and finally solve every target by a finite-dimensional fixed-point argument.

4.1 Frozen and state-dependent shifts

First ignore the dependence on s . The identity $wv = 1$ supplies an inverse for any independently frozen choice of the pairs.

Lemma 4.1 (Frozen shift). *Choose for every (i, n) a pair $v^{i,n}, w^{i,n}$ satisfying (W1), independently between blocks. Define $S : D \rightarrow U$ by*

$$a_{i,0} = t_i, \quad a_{i,n} = w^{i,n} x^{i,n}, \quad (Sx)^{i,n} = x^{i,n} + v^{i,n}(a_{i,n-1} - a_{i,n}). \quad (4.1)$$

Then S is a bounded linear bijection with $\|S\| \leq 5$ and $\|S^{-1}\| \leq 5$.

Proof. The row bounds give $|a_{i,n-1} - a_{i,n}| \leq 4r_{i,n}(x)$; for $n = 1$, the predecessor is just t_i , and the same bound holds. Thus

$$\|(Sx)^{i,n}\|_\infty \leq 5r_{i,n}(x).$$

By (3.1), the block norms tend to zero, so $Sx \in U$ and $\|S\| \leq 5$.

For $y \in U$, put $b_{i,n} = w^{i,n} y^{i,n}$ and define

$$t_i = b_{i,1}, \quad x^{i,n} = y^{i,n} - v^{i,n}(b_{i,n} - b_{i,n+1}). \quad (4.2)$$

The block norms of y tend to zero off finite sets: otherwise infinitely many coordinates of y would stay away from zero. Since $|b_{i,n}| \leq 2\|y^{i,n}\|_\infty$, the scalar family $(b_{i,n})$ belongs to $c_0(\mathbb{N}^2)$. Formula (4.2) therefore gives $t \in E$ and block norms of x tending to zero. Because the blocks are finite, this is $x \in D$. It also gives $\|x\|_D \leq 5\|y\|_U$.

Applying $w^{i,n}$ to (4.2) yields $w^{i,n} x^{i,n} = b_{i,n+1}$, so (4.1) gives $Sx = y$. Conversely, when $y = Sx$, applying $w^{i,n}$ to (4.1) gives $b_{i,n} = a_{i,n-1}$; (4.2) then recovers t_i and $x^{i,n}$. The formulas are mutual inverses and prove the inverse bound. $\square$

For $s \in B$, let S_s be the frozen shift using $v^\lambda(s), w^\lambda(s)$ in every block. Define

$$g(0) = 0, \quad g(x) = S_{s_x} x, \quad s_x = \frac{x}{\|x\|_D} \quad (x \neq 0). \quad (4.3)$$

This map is positively homogeneous. Its weights are recomputed from the input, so Lemma 4.1 alone does not invert g .

Proposition 4.2 (Global distance bounds). *For every $x, x' \in D$,*

$$\left(\frac{1}{5} - 24\eta\right) \|x - x'\|_D \leq \|g(x) - g(x')\|_U \leq (5 + 24\eta) \|x - x'\|_D. \quad (4.4)$$

In particular, g is injective and continuous.

Proof. Suppose first that $x, x' \neq 0$. Put $d = \|x - x'\|_D$ and $M = \max\{\|x\|_D, \|x'\|_D\}$. If $\|x\|_D = M$, splitting $x/M - x'/\|x'\|$ at x'/M , and using $\|x\| - \|x'\| \leq d$, gives

$$\|s_x - s_{x'}\|_D \leq \frac{2d}{M}. \quad (4.5)$$

The other order is the same.

At each level λ , freeze the entire pair $(\bar{v}^\lambda, \bar{w}^\lambda)$ at whichever of $s_x, s_{x'}$ has larger r_λ , breaking a tie arbitrarily. These choices may vary with λ , but Lemma 4.1 still applies to the resulting $\bar{S}$. For $e = x$ or x' , the chosen pair either agrees with the weights at s_e or the radius at s_e is the smaller one. Thus (W2) and (4.5) give

$$r_\lambda(e) \max\{\|v^\lambda(s_e) - \bar{v}^\lambda\|_\infty, \|w^\lambda(s_e) - \bar{w}^\lambda\|_1\} \leq 2\eta d. \quad (4.6)$$

This remains valid when $r_\lambda(e) = 0$.

In block $\lambda = (i, n)$, first change its vector v^λ while keeping the two scalars computed with the original rows. Their difference has modulus at most $4r_\lambda(e)$, so (4.6) bounds this change by $8\eta d$. Changing the current row changes its scalar by at most $2\eta d$, since $\|e^\lambda\|_\infty \leq r_\lambda(e)$. If $n > 1$, changing the predecessor row costs another $2\eta d$, using (4.6) at $(i, n-1)$ and $\|e^{i, n-1}\|_\infty \leq r_{i, n-1}(e)$. For $n = 1$, the predecessor t_i does not change. Since $\|\bar{v}^\lambda\|_\infty \leq 1$, we have

$$\|(S_{s_e} e - \bar{S} e)^\lambda\|_\infty \leq 12\eta d. \quad (4.7)$$

Taking the supremum over blocks and adding the two endpoint errors,

$$\|g(x) - g(x') - \bar{S}(x - x')\|_U \leq 24\eta d.$$

The two frozen bounds give $\frac{1}{5}d \leq \|\bar{S}(x - x')\|_U \leq 5d$, proving (4.4). If one endpoint is zero, Lemma 4.1 applied to $S_{s_x} x$ gives the stronger constants $1/5$ and 5 . Finally, $1/5 - 24\eta > 0$ for $\eta = 10^{-3}$. $\square$

4.2 Terminal scalars and Brouwer surjectivity

The lower bound in (4.4) proves injectivity and will prove that the image is closed. It does not prove that every target has a preimage. For that, we solve finite-coordinate targets with an extra terminal scalar in each row.

Proposition 4.3 (Onto-ness of the nonlinear shift). *The map $g : D \rightarrow U$ in (4.3) is onto. Together with Proposition 4.2, it is a bi-Lipschitz bijection.*

Proof. For $N \geq 1$, let U_N consist of the blocks with $1 \leq i, n \leq N$, all other coordinates zero. Let D_N consist of these blocks and $t_1, \dots, t_N$, all other input coordinates zero. They are finite-dimensional, and

$$\dim D_N = N + \sum_{i, n \leq N} |\Gamma_{i, n}| = \dim(U_N \oplus_\infty \mathbb{R}^N). \quad (4.8)$$

For $s \in B$, set

$$S_s^{(N)} : D_N \rightarrow U_N \oplus_\infty \mathbb{R}^N, \quad S_s^{(N)} x = \left((S_s x)^{i, n} \right)_{i, n \leq N}, (w^{i, N}(s) x^{i, N})_{i \leq N}. \quad (4.9)$$

The last N coordinates retain the terminal scalars $a_{i, N}$.

For clarity, this finite map has an explicit inverse. Given $(y, (c_i)_{i \leq N})$, define

$$b_{i, n} = w^{i, n}(s) y^{i, n}, \quad a_{i, 0} = b_{i, 1}, \quad a_{i, n} = b_{i, n+1} \quad (n < N), \quad a_{i, N} = c_i,$$

then put

$$t_i = a_{i, 0}, \quad x^{i, n} = y^{i, n} - v^{i, n}(s)(a_{i, n-1} - a_{i, n}) \quad (n \leq N). \quad (4.10)$$

Because $a_{i, n-1} = b_{i, n}$, applying $w^{i, n}(s)$ to (4.10) gives $w^{i, n}(s) x^{i, n} = a_{i, n}$. Substitution in (4.9) returns (y, c) . Conversely, applying the rows to (4.9) recovers the $b_{i, n}$, and (4.10) recovers every input coordinate. Thus (4.10) is the inverse.

The same estimates as for the frozen shift give

$$\|S_s^{(N)}\| \leq 5, \quad \|(S_s^{(N)})^{-1}\| \leq 5. \quad (4.11)$$

For the inverse bound, if $h_0 = \|(y, c)\|$, then $|b_{i,n}| \leq 2h_0$, all $a_{i,n}$ have modulus at most $2h_0$, and (4.10) has block norm at most $h_0 + 4h_0$. For the forward bound, the block estimate is $5\|x\|_D$ and the terminal coordinates are at most $2\|x\|_D$. The finitely many weights in (4.10) are continuous in s . Hence $s \mapsto (S_s^{(N)})^{-1}$ is continuous as a map into the linear operators between these finite-dimensional spaces.

Figure 1 displays the terminal coordinate and the single possible outgoing block of a finite row.

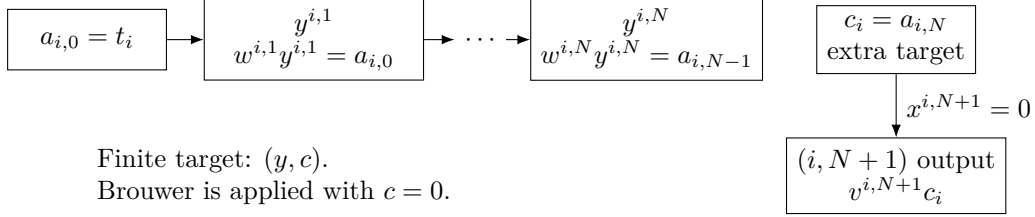


Figure 1: A row of length N with frozen weights and independent extra target scalar $c_i = a_{i,N}$. The block readouts recover the predecessor scalars, while c_i records the scalar not read by any block in the finite row. At the nonlinear fixed point it is set to zero, which kills the only possible output spillover into the next block.

Let $0 \neq y \in U_N$, and set $h = \|y\|_U > 0$. On the closed ball $\{x \in D_N : \|x\|_D \leq 5h\}$, define

$$s(x) = \frac{x}{\max\{\|x\|_D, h/6\}}, \quad T(x) = (S_{s(x)}^{(N)})^{-1}(y, 0). \quad (4.12)$$

The denominator is positive even at $x = 0$, so $s(x) \in B$ depends continuously on x . Equations (4.10)–(4.11) show that T is a continuous self-map of this finite-dimensional closed ball. Brouwer's fixed-point theorem gives $x = T(x)$. At that fixed point,

$$h = \|(y, 0)\| = \|S_{s(x)}^{(N)}x\| \leq 5\|x\|_D,$$

and therefore $\|x\|_D \geq h/5 > h/6$. The state in (4.12) is now the normalized state s_x , not the regularized state used near zero. The blocks of $g(x)$ in the N -square are y , and (4.9) gives $a_{i,N} = 0$ for every $i \leq N$.

It remains to check the blocks omitted from the finite equation. The input is zero outside D_N . The only outside block with a possibly nonzero predecessor scalar is $(i, N + 1)$ for $i \leq N$. There

$$(g(x))^{i,N+1} = v^{i,N+1}(s_x)a_{i,N} = 0.$$

All other outside blocks have zero input and zero predecessor scalar. Thus $g(x) = y$ in U . The zero target is $g(0)$, so $U_N \subset g(D)$ for every N .

Every finitely supported element of U belongs to some U_N , and such elements are dense. The image $g(D)$ is closed: if $g(x_k)$ converges in U , the positive lower bound (4.4) makes (x_k) Cauchy in the Banach space D ; its limit x satisfies $g(x_k) \rightarrow g(x)$ by continuity. Hence $g(D)$ is both dense and closed in U , and is all of U . $\square$

Let $P : D \rightarrow U$ be the projection and, for $s \in B$, put

$$h(s) = (S_s - P)s. \quad (4.13)$$

The frozen lemma ensures $h(s) \in U$, and (3.5) is exactly its γ -coordinate. Notice that $S_s s$ uses s itself as state even when $\|s\| < 1$, whereas $g(s)$ uses $s/\|s\|$. The lifting argument will use (4.13) on B and (4.3) only on normalized states. By (3.6), the correction coordinates have common bounds

$$A := \sup_{\gamma} \|f_{\gamma}\|_{\infty} \leq 4, \quad L_0 := \sup_{\gamma} \text{Lip}(f_{\gamma}) \leq 4 + 12\eta. \quad (4.14)$$

5 Selected tests and their supports

The next goal is to construct a Banach space that lifts the correction $h(s) = (S_s - P)s$ and contains no linear c_0 . We use the coordinate functions f_{γ} of (3.5), with exactly the finite-dependence and support assertions (3.6)–(3.7). The operations below enlarge the family of tests while retaining quantitative control of where each test can be nonzero.

5.1 Two local operations

Fix a label $\gamma = (\lambda, j, \xi)$ and its nonempty finite set $I = I_{\gamma}$. A function on B depending only on I is identified with its function on the cube $[-1, 1]^I$, by evaluating at the zero extension of a cube point. Write $s|_I$ for coordinate restriction. For a dyadic scale $\varepsilon = 2^{-a}$, $a \in \mathbb{N}_0 = \{0, 1, \dots\}$, let

$$\mathcal{Q}_{I,\varepsilon} = ([-1, 1] \cap (\varepsilon/8)\mathbb{Z})^I.$$

This is a finite grid including both endpoints in each coordinate. Every point of the cube is within $\varepsilon/16$, hence within $\varepsilon/8$, of a grid point in the supremum norm.

Define the 2-Lipschitz function

$$\chi(t) = \begin{cases} 0, & 0 \leq t \leq 1/2, \\ 2t - 1, & 1/2 < t < 1, \\ 1, & t \geq 1. \end{cases}$$

For a Lipschitz function u depending on I , and $q \in \mathcal{Q}_{I,\varepsilon}$, put

$$(G_{q,\varepsilon}u)(s) = u(q) + \chi\left(\frac{\|s|_I - q\|_{\infty}}{\varepsilon}\right) (u(s) - u(q)), \quad (5.1)$$

$$R_{q,\varepsilon}u = G_{q,\varepsilon}u - u. \quad (5.2)$$

Here $u(q)$ means evaluation at the zero extension of q . Both operations are linear in u , retain dependence on I , and $G_{q,\varepsilon}u$ is the constant $u(q)$ on the open cylinder $\|s|_I - q\|_{\infty} < \varepsilon/2$. Thus G makes u locally constant and changes it only inside the radius- ε cylinder; R records that change. The identity $G = \text{Id} + R$ will separate fixed parts of a test from changes occurring at small scales.

Lemma 5.1 (Bounds and geometric support). *For every such u, q, ε ,*

$$\|R_{q,\varepsilon}u\|_{\infty} \leq \varepsilon \text{Lip}(u), \quad \text{Lip}(R_{q,\varepsilon}u) \leq 3 \text{Lip}(u), \quad \text{Lip}(G_{q,\varepsilon}u) \leq 4 \text{Lip}(u), \quad (5.3)$$

$$\|G_{q,\varepsilon}u\|_{\infty} \leq \|u\|_{\infty}, \quad \|R_{q,\varepsilon}u\|_{\infty} \leq 2\|u\|_{\infty}. \quad (5.4)$$

Let

$$K_I(u) = \overline{\{s|_I : s \in B, u(s) \neq 0\}}^{[-1,1]^I}.$$

In the next display write $G = G_{q,\varepsilon}$ and $R = R_{q,\varepsilon}$. With $K_{\varepsilon} = \{p : \text{dist}_{\infty}(p, K) \leq \varepsilon\}$ and $\emptyset_{\varepsilon} = \emptyset$, one has

$$K_I(Gu), K_I(Ru) \subset K_I(u)_{\varepsilon}, \quad K_I(Ru) \subset \overline{B}_{\infty}(q, \varepsilon). \quad (5.5)$$

Proof. Set $a(s) = 1 - \chi(\|s|_I - q\|_\infty/\varepsilon)$. Then

$$R_{q,\varepsilon}u(s) = a(s)(u(q) - u(s)), \quad 0 \leq a \leq 1, \quad \text{Lip}(a) \leq 2/\varepsilon.$$

If $a(s) \neq 0$, then $\|s|_I - q\|_\infty < \varepsilon$; the cylindrical Lipschitz bound gives $|u(q) - u(s)| \leq \varepsilon \text{Lip}(u)$. This proves the first supremum estimate in (5.3).

To compare $Ru(s)$ and $Ru(t)$, there is nothing to prove if both values of a are zero. Otherwise interchange s, t , if needed, so that $a(t) \neq 0$. The identity

$$Ru(s) - Ru(t) = a(s)(u(t) - u(s)) + (a(s) - a(t))(u(q) - u(t))$$

bounds the two terms by $\text{Lip}(u)\|s-t\|_D$ and $(2/\varepsilon)\|s-t\|_D \varepsilon \text{Lip}(u)$, respectively. Thus $\text{Lip}(Ru) \leq 3 \text{Lip}(u)$, and $Gu = u + Ru$ gives $\text{Lip}(Gu) \leq 4 \text{Lip}(u)$. The formula for G is a convex combination of $u(q)$ and $u(s)$, so it does not increase the supremum norm. The bound for $R = G - u$ follows.

For the support assertions, if a new output is nonzero at s while $u(s) = 0$, then $u(q) \neq 0$ and $\|s|_I - q\|_\infty < \varepsilon$. If $u(s) \neq 0$, its projection already lies in $K_I(u)$. This covers both G and R . Taking closures gives their inclusion in $K_I(u)_\varepsilon$; the latter set is closed because the cube is compact. In addition, $Ru(s) = 0$ whenever $\|s|_I - q\|_\infty \geq \varepsilon$, which proves its ball inclusion. If $u = 0$, all statements are immediate. $\square$

5.2 Scale budgets and tail factors

For $h \geq 1$, let $\mathcal{T}_h$ consist of functions obtained from a single f_γ , $\gamma = (\lambda, j, \xi)$, by $m \leq h$ successive operations G or R on I_γ . A representation with $m = 0$ is allowed. For $m \geq 1$, the scales in their order of application must satisfy

$$\varepsilon_1 \geq \varepsilon_2 \geq \cdots \geq \varepsilon_m > 0, \quad \sum_{k=1}^m \varepsilon_k < \frac{\ell_j}{2}. \quad (5.6)$$

The sum is zero for $m = 0$. Each scale is dyadic and each center is in its corresponding finite grid. We refer to γ as a representing label, since a function may have more than one representation. The strict inequality leaves a positive amount of budget for a further sufficiently small operation. The order will ensure that every operation after a small one has at most that scale.

For a continuous function on B , let $\text{supp}_B u = \overline{\{s \in B : u(s) \neq 0\}}^B$. All support statements refer to these concrete generated functions. Later, molecule pairings will ignore additive constants, but we do not replace a generated function by another representative modulo constants: that could change its support and its constant value on a cylinder.

Lemma 5.2 (Propagation of the correction support). *Suppose $u \in \mathcal{T}_h$ has a representation based at $\gamma = (\lambda, j, \xi)$ with total scale $e = \sum_{k=1}^m \varepsilon_k$. On $\text{supp}_B u$,*

$$\begin{cases} \|\pi_j s - \xi\|_\infty \leq 2d_j + e, & j \geq 1, \\ r_\lambda(s) \geq \ell_j - e, & j < J_\lambda. \end{cases} \quad (5.7)$$

In particular, on the same support,

$$\begin{cases} \|\pi_j s - \xi\|_\infty \leq 2d_j + \ell_j/2, & j \geq 1, \\ r_\lambda(s) \geq \ell_j/2, & j < J_\lambda. \end{cases} \quad (5.8)$$

Proof. Start with (3.7), which holds on the relative support of f_γ as well as its nonzero set, because its two inequalities describe closed subsets of B . Suppose the claim has been proved for the old function and add an operation of scale ε . If the new function is nonzero at s and the old one is zero there, the proof of Lemma 5.1 gives an old nonzero value at the zero extension of

q , with $\|s|_{I_\gamma} - q\|_\infty < \varepsilon$. Otherwise the old support already contains s . Both π_j and r_λ depend only on I_γ ; coordinate restriction has norm one, and r_λ is 1-Lipschitz on that cube. The first upper bound therefore increases by at most ε , and the second lower bound decreases by at most ε . Induction proves (5.7) on the nonzero set. Taking relative closure preserves the closed inequalities. Finally, (5.6) gives (5.8). $\square$

For $N \geq 1$, define the coordinate tail

$$\tau_N(s) = \sup_{k > N} |s_k|, \quad b_{c,N}(s) = \min\{1, (2 - \tau_N(s)/c)_+\} \quad (c > 0), \quad (5.9)$$

where $a_+ = \max\{a, 0\}$. The function τ_N is 1-Lipschitz, and $\tau_N(s) \rightarrow 0$ for each $s \in B$. Both $b_{c,N}$ and $b_{c,N} - 1$ have supremum norm at most one and Lipschitz constant at most $1/c$. Their relative supports satisfy

$$\text{supp}_B b_{c,N} \subset \{\tau_N \leq 2c\}, \quad \text{supp}_B (b_{c,N} - 1) \subset \{\tau_N \geq c\}. \quad (5.10)$$

For example, the first function vanishes when $\tau_N \geq 2c$, and the second vanishes when $\tau_N \leq c$; closure gives the displayed weak inequalities.

Let $\mathcal{V}_h$ consist of all products

$$u \prod_{r=1}^l \beta_r \quad (5.11)$$

where $u \in \mathcal{T}_h$ has a representing label with $j \leq h$, $0 \leq l \leq h$, and each β_r is b_{c_r, N_r} or $b_{c_r, N_r} - 1$, with $N_r \geq 1$ and

$$c_r \in \{2^{-a} : 0 \leq a \leq h\}.$$

The empty product is one. The support of a product is contained in the support of its cylindrical factor u , so every such test inherits (5.8). We will use the countable list of classes

$$\mathcal{T}_1, \mathcal{V}_1, \mathcal{T}_2, \mathcal{V}_2, \dots \quad (5.12)$$

Lemma 5.3 (Bounds in each fixed class). *With A, L_0 from (4.14), the following bounds are uniform over all tests of the indicated class:*

$u \in \mathcal{T}_h$	$\ u\ _\infty$	$\text{Lip}(u)$	(5.13)
$u \in \mathcal{V}_h$	$2^h A$	$4^h L_0$	
$u \in \mathcal{V}_h$	$2^h A$	$4^h L_0 + h4^h A$	

Proof. Iterating (5.3)–(5.4) gives the first row. For the second, every tail factor has supremum norm at most one and Lipschitz constant at most 2^h . A product of at most h such factors has Lipschitz constant at most $h2^h$, by telescoping the product difference. The product rule then gives

$$\text{Lip}\left(u \prod_r \beta_r\right) \leq 4^h L_0 + (2^h A)(h2^h) = 4^h L_0 + h4^h A.$$

Its supremum norm is at most $2^h A$. $\square$

We have two useful consequences of the support bounds. For a fixed bounded set of bands j , the second line of (5.8) gives a positive radial threshold whenever $j < J_\lambda$. For $j \rightarrow \infty$, the first line confines the first j coordinates to a radius $2d_j + \ell_j/2 \rightarrow 0$. Section 7 will use precisely these alternatives, after the next section extends the tests from finite molecules to the completed Banach space.

6 Molecules and the lifted correction

We now turn the selected tests into a Banach space. The norm must do two jobs: control point differences so that the correction can be lifted Lipschitzly, and remember lists of tests whose signed sums have a common Lipschitz bound. The latter feature will rule out linear c_0 .

6.1 The quotient and its completion

Let $\mathcal{M}_0(B)$ be the real vector space of mass-zero finite formal sums

$$m = \sum_{s \in B} a_s \delta_s, \quad \#\{s : a_s \neq 0\} < \infty, \quad \sum_s a_s = 0.$$

For a real function u on B , set $\langle u, m \rangle = \sum_s a_s u(s)$. Constants pair to zero. This follows the mass-zero molecule construction of Arens and Eells [4] and its later Lipschitz-free formulation [5]. The norm below is built from the selected classes (5.12); no universal property of the usual free space is assumed.

For every triple $d = (\mathcal{L}, C, p)$, where $\mathcal{L}$ is one of the listed classes, $C \in \mathbb{N}$, and $p = 1 + 1/k$ for $k \in \mathbb{N}$, define

$$\sigma_d(m) = \sup \left\{ \left(\sum_{\nu=1}^l |\langle u_\nu, m \rangle|^p \right)^{1/p} : \begin{array}{l} l \geq 0, \quad u_1, \dots, u_l \in \mathcal{L}, \\ \text{Lip} \left(\sum_{\nu=1}^l e_\nu u_\nu \right) \leq C \text{ for every } (e_\nu) \in \{-1, 1\}^l \end{array} \right\}. \quad (6.1)$$

The empty list contributes zero. Lists may contain repetitions and tests with the same representing label; their sole compatibility condition is the displayed Lipschitz bound for every choice of signs. Equivalently, a list is admissible precisely when

$$\sum_{\nu=1}^l |u_\nu(s) - u_\nu(t)| \leq C \|s - t\|_D \quad (s, t \in B). \quad (6.2)$$

To obtain this inequality, choose signs matching the increments $u_\nu(s) - u_\nu(t)$; the converse follows from the triangle inequality.

Enumerate the triples by $d = 1, 2, \dots$, write C_d for their integer component, and set

$$\alpha_d = \frac{2^{-d}}{C_d^2}, \quad \|m\|_0 = \left(\sum_{d \geq 1} \alpha_d \sigma_d(m)^2 \right)^{1/2}. \quad (6.3)$$

The outer square sum and the exponents approaching one serve distinct purposes in Section 7. The square sum forces one fixed seminorm to detect a subsequence of any hypothetical c_0 basis. For that fixed test class and Lipschitz bound, a second exponent $p' < p$ converts detection by a list into detection by one scalar test. Both choices are part of the obstruction, as well as of the norm construction.

Lemma 6.1 (Exact completed norm). *Formula (6.3) is a finite seminorm on $\mathcal{M}_0(B)$. Let N be its kernel, and let Z be the Banach completion of $\mathcal{M}_0(B)/N$ with the induced norm. Then Z is separable, and*

$$\zeta : B \rightarrow Z, \quad \zeta(s) = [\delta_s - \delta_0] \quad (6.4)$$

is 1-Lipschitz with $\zeta(0) = 0$. Each σ_d descends and extends continuously to a seminorm on Z , and the exact formula

$$\|z\|_Z^2 = \sum_{d \geq 1} \alpha_d \sigma_d(z)^2 \quad (6.5)$$

holds for every $z \in Z$.

Proof. For $s, t \in B$ and an admissible list for d , inequality (6.2) bounds the sum of the absolute increments. Since the ℓ_p norm is no larger than this sum,

$$\sigma_d(\delta_s - \delta_t) \leq C_d \|s - t\|_D. \quad (6.6)$$

For a fixed list, Minkowski's inequality makes its expression in (6.1) a seminorm in m ; the supremum of these expressions is also a seminorm, once finite. Every $m = \sum_s a_s \delta_s$ of mass zero equals $\sum_s a_s (\delta_s - \delta_0)$. Thus

$$\sigma_d(m) \leq C_d \sum_s |a_s| \|s\|_D.$$

This proves finiteness both of σ_d and of $\|m\|_0$, because $\sum_d \alpha_d C_d^2 = \sum_d 2^{-d} = 1$. Minkowski's inequality in ℓ_2 , applied to the weighted seminorms, proves subadditivity of $\|\cdot\|_0$; homogeneity is immediate. Its kernel is therefore a linear subspace, and quotienting by it produces a norm.

Equation (6.6) and the same square sum give

$$\|\delta_s - \delta_t\|_0 \leq \|s - t\|_D.$$

This proves the assertion about ζ . The ball B is separable because D is a countable-coordinate c_0 space. If B_0 is a countable dense subset, continuity of ζ makes $\zeta(B_0)$ dense in $\zeta(B)$. The real linear span of $\zeta(B)$ is the quotient molecule space; its rational span over B_0 is consequently dense in Z . Hence Z is separable.

For molecules m, m' ,

$$|\sigma_d(m) - \sigma_d(m')| \leq \sigma_d(m - m') \leq \alpha_d^{-1/2} \|m - m'\|_0. \quad (6.7)$$

In particular σ_d vanishes on N , descends, and extends continuously to Z ; subadditivity and homogeneity pass to limits. Moreover the map

$$m \mapsto (\sqrt{\alpha_d} \sigma_d(m))_{d \geq 1}$$

is 1-Lipschitz into ℓ_2 , since the coordinate differences are bounded by $\sqrt{\alpha_d} \sigma_d(m - m')$. If molecules converge to $z \in Z$, their images therefore converge in ℓ_2 . The d -th coordinate of the limit is $\sqrt{\alpha_d} \sigma_d(z)$ by continuity of each seminorm. The norms of those images are exactly the molecule norms, which converge to $\|z\|_Z$. This proves (6.5). $\square$

6.2 Fixed-class duality

The obstruction argument will modify a test through a pointwise limit after a vector in Z has already been fixed. The next lemma justifies that passage, and also retains the exact list formula after completion.

Lemma 6.2 (Dual tests and pointwise limits). *Every test in $\mathcal{T}_h$ or $\mathcal{V}_h$ defines a bounded linear functional on Z by its molecule pairing. For each fixed class $\mathcal{L}$, there is a finite common bound $B_{\mathcal{L}}$ for these dual norms. For each fixed triple $d = (\mathcal{L}, C, p)$, formula (6.1) remains exact on Z , with the extended test pairings.*

If $u_n \in \mathcal{L}$ and $u_n(s) \rightarrow u(s)$ for every $s \in B$, then the pointwise limit defines a bounded functional on Z of norm at most $B_{\mathcal{L}}$, and $\langle u_n, z \rangle \rightarrow \langle u, z \rangle$ for each fixed $z \in Z$. More precisely, for any molecule class m and any $z \in Z$,

$$|\langle u_n - u, z \rangle| \leq |\langle u_n - u, m \rangle| + 2B_{\mathcal{L}} \|z - m\|_Z. \quad (6.8)$$

Proof. Choose an integer $C_{\mathcal{L}}$ no smaller than the uniform Lipschitz bound in Lemma 5.3. Every singleton (u) , $u \in \mathcal{L}$, is admissible for the one triple $d_{\mathcal{L}} = (\mathcal{L}, C_{\mathcal{L}}, 2)$. Thus

$$|\langle u, m \rangle| \leq \sigma_{d_{\mathcal{L}}}(m) \leq \alpha_{d_{\mathcal{L}}}^{-1/2} \|m\|_0. \quad (6.9)$$

It vanishes on N and extends to Z^* , uniformly with $B_{\mathcal{L}} = \alpha_d^{-1/2}$.

For an admissible list $\mathbf{u} = (u_1, \dots, u_l)$ of a fixed triple d , its evaluation map $\Lambda_{\mathbf{u}}m = (\langle u_\nu, m \rangle)_{\nu=1}^l$ into ℓ_p^l has norm at most $\alpha_d^{-1/2}$ on the molecule quotient. It therefore extends with that bound to Z . The supremum of $\|\Lambda_{\mathbf{u}}z\|_p$ over all admissible lists is finite and $\alpha_d^{-1/2}$ -Lipschitz as a seminorm in z : its change between z, z' is bounded by the same supremum at $z - z'$. This supremum agrees with σ_d on the dense molecule quotient. It is therefore the continuous extension of σ_d , which proves the exact formula on Z .

For a pointwise-convergent sequence, the pairings converge on every finite molecule. Inequality (6.9) passes to that pointwise limit, so its molecule functional extends with norm at most $B_{\mathcal{L}}$. Applying the two common dual bounds to $z - m$ gives (6.8). For a prescribed z and tolerance $\delta > 0$, choose a molecule class m with $\|z - m\|_Z < \delta/(4B_{\mathcal{L}})$, and then choose n so large that the finite pairing in (6.8) is less than $\delta/2$. This proves convergence on z . The choice of m and n may depend on z ; no convergence uniform over a moving sequence of vectors is asserted. $\square$

Lemma 6.3 (The c_0 -valued correction map). *There is a bounded linear map $Q : Z \rightarrow U$ such that*

$$Q\zeta(s) = h(s) = (S_s - P)s \quad (s \in B). \quad (6.10)$$

Proof. Define on formal molecules

$$Q_0m = \sum_s a_s h(s) \in U.$$

This finite sum lies in U , not merely in ℓ_∞ , because each $h(s)$ lies in U . Each coordinate function f_γ belongs to $\mathcal{T}_1$ through its zero-operation representation. Choose an integer $C_0 \geq L_0$ and the triple $d_0 = (\mathcal{T}_1, C_0, 2)$. Every singleton (f_γ) is admissible, whence

$$\|Q_0m\|_U = \sup_\gamma |\langle f_\gamma, m \rangle| \leq \sigma_{d_0}(m) \leq \alpha_{d_0}^{-1/2} \|m\|_0.$$

Thus Q_0 vanishes on N , descends, and extends to a bounded linear map into the complete space U . Since $h(0) = 0$, applying it to $\delta_s - \delta_0$ proves (6.10). $\square$

6.3 Radial extension

The point map ζ is defined on B . Its radial extension lifts the homogeneous shift correction on all of D .

Lemma 6.4 (Lipschitz lift). *Define*

$$K(0) = 0, \quad K(x) = \|x\|_D \zeta\left(\frac{x}{\|x\|_D}\right) \quad (x \neq 0). \quad (6.11)$$

Then $K : D \rightarrow Z$ is positively homogeneous and $\text{Lip}(K) \leq 3$. It satisfies the exact identity

$$QK = g - P. \quad (6.12)$$

Proof. Since $\zeta(0) = 0$ and $\text{Lip}(\zeta) \leq 1$, $\|\zeta(s)\|_Z \leq 1$ for $s \in B$. Suppose $a = \|x\|_D \geq b = \|x'\|_D > 0$, and write $s = x/a$, $s' = x'/b$. Splitting at $b\zeta(s)$ gives

$$\|K(x) - K(x')\|_Z \leq (a - b)\|\zeta(s)\|_Z + b\|\zeta(s) - \zeta(s')\|_Z \leq \|x - x'\|_D + b\|s - s'\|_D.$$

The normalization estimate (4.5) bounds the last term by $2\|x - x'\|_D$. If an endpoint is zero, use $\|K(x)\|_Z \leq \|x\|_D$. This proves the Lipschitz bound. For $x \neq 0$, (6.10), (4.13), and (4.3) give

$$QK(x) = \|x\|_D h(x/\|x\|_D) = S_{x/\|x\|_D}x - Px = g(x) - Px.$$

All terms vanish at zero, and positive homogeneity follows directly from (6.11). $\square$

We have now obtained the exact data Q, K, g of Lemma 2.1. No surjectivity of Q was used or proved. The remaining task is to show that the completed space Z has no linear copy of c_0 .

7 Detectors exclude linear c_0

A linear copy of c_0 would provide vectors bounded away from zero whose finite signed sums are uniformly bounded. We first show that one class of tests detects a subsequence of these vectors. We then identify the additional property needed for a contradiction: the tests' finite signed sums must have one Lipschitz bound. The rest of the section obtains this property by modifying the detectors and separating or nesting their supports. Supports always refer to the concrete generated functions on B .

7.1 From bounded signed sums to individual detectors

The extraction has two stages. The outer square sum prevents detection from escaping through different seminorms; within one seminorm, the exponents approaching one prevent it from spreading over arbitrarily many small scalar pairings.

Lemma 7.1 (One class detects a subsequence). *Suppose $(m_i) \subset Z$ satisfies, for some $r, M > 0$,*

$$\|m_i\|_Z \geq r, \quad \left\| \sum_{i \in H} \epsilon_i m_i \right\|_Z \leq M \quad \text{for all finite } H \text{ and all signs } \epsilon_i. \quad (7.1)$$

Then $u(m_i) \rightarrow 0$ for every $u \in Z^$. After a subsequence, there are tests F_i in one listed class $\mathcal{L}$ and a constant $\delta > 0$ with*

$$|\langle F_i, m_i \rangle| \geq \delta. \quad (7.2)$$

Proof. For any fixed $u \in Z^*$, choosing signs matching $u(m_i)$ in (7.1) gives

$$\sum_{i \in H} |u(m_i)| \leq M \|u\|$$

for every finite H . Thus $\sum_i |u(m_i)| < \infty$, and in particular $u(m_i) \rightarrow 0$. Also $\|m_i\| \leq M$.

We first prove that one fixed seminorm stays away from zero on a subsequence. Otherwise $\sigma_d(m_i) \rightarrow 0$ for every d . The vectors

$$v_i = (\sqrt{\alpha_d} \sigma_d(m_i))_{d \geq 1}$$

then converge coordinatewise to zero in ℓ_2 , while $r \leq \|v_i\|_2 = \|m_i\|_Z \leq M$. A direct gliding-hump selection gives disjoint finite intervals A_i of indices, after a subsequence, such that

$$\sum_{d \in A_i} \alpha_d \sigma_d(m_i)^2 \geq r^2/4. \quad (7.3)$$

Indeed, once the endpoint of the preceding interval is fixed, coordinatewise convergence makes the squared mass before that endpoint less than $r^2/4$ for a sufficiently late v_i . Then choose a new finite endpoint so that the squared tail mass is less than $r^2/4$. The mass in between is at least $r^2/2$, which implies (7.3).

The function σ_d^2 is even and convex, because a seminorm is convex and squaring is convex and increasing on $[0, \infty)$. For fixed y, m , Jensen's inequality applied to $y + m$ and $-y + m$ gives

$$\frac{\sigma_d(y + m)^2 + \sigma_d(y - m)^2}{2} \geq \sigma_d(m)^2. \quad (7.4)$$

For independent random signs and a finite H , average first over the sign of m_i and use (7.4). Thus

$$\mathbb{E} \sigma_d \left(\sum_{k \in H} \epsilon_k m_k \right)^2 \geq \sigma_d(m_i)^2 \quad (i \in H).$$

The exact norm formula, (7.1), the disjointness of the A_i , and (7.3) imply

$$M^2 \geq \mathbb{E} \left\| \sum_{k \in H} \epsilon_k m_k \right\|_Z^2 \geq \sum_{i \in H} \sum_{d \in A_i} \alpha_d \sigma_d(m_i)^2 \geq |H| r^2 / 4,$$

a contradiction for large H . Hence some $d = (\mathcal{L}, C, p)$ has $\sigma_d(m_i) \geq a > 0$ on a subsequence.

Choose a listed exponent $p' = 1 + 1/k' < p$, with the same $\mathcal{L}, C$, and call its triple d' . Its seminorm is bounded on these vectors by

$$\sigma_{d'}(m_i) \leq \alpha_{d'}^{-1/2} M =: B_0.$$

For each i , choose a finite admissible list for d whose ℓ_p value on m_i is at least $a/2$. Put $x_\nu = |\langle u_\nu, m_i \rangle|$. The same list is admissible for d' , because only the exponent changed. Consequently

$$(a/2)^p \leq \sum_{\nu} x_{\nu}^p \leq (\max_{\nu} x_{\nu})^{p-p'} \sum_{\nu} x_{\nu}^{p'} \leq (\max_{\nu} x_{\nu})^{p-p'} B_0^{p'}.$$

The list is nonempty and $B_0 > 0$. One of its tests therefore has evaluation at least

$$\delta = \left(\frac{(a/2)^p}{B_0^{p'}} \right)^{1/(p-p')} > 0.$$

Choosing that test as F_i proves (7.2). □

7.2 The growth forced by compatible detectors

Call a sequence of tests *compatible* if all its finite signed sums have a common Lipschitz bound. The norm was defined using precisely such lists. Once compatible tests detect the vectors, a single seminorm forces the following growth.

Lemma 7.2 (Growth from compatible detectors). *Let $(m_i) \subset Z$, and let F_i belong to one listed class $\mathcal{L}$. Suppose there are $\delta > 0$ and an integer $C \geq 1$ such that*

$$|\langle F_i, m_i \rangle| \geq \delta, \quad \text{Lip} \left(\sum_{i \in H} \epsilon_i F_i \right) \leq C$$

for every i , every finite H , and all signs ϵ_i . For the listed triple $d = (\mathcal{L}, C, 2)$, every nonempty finite H satisfies

$$\max_{\epsilon_i \in \{-1, 1\}} \left\| \sum_{i \in H} \epsilon_i m_i \right\|_Z \geq \sqrt{\alpha_d} \delta \sqrt{|H|}. \quad (7.5)$$

Proof. The list $(F_k)_{k \in H}$ is admissible for d , including when labels repeat. For independent real random signs, the exact list formula in Lemma 6.2 gives

$$\begin{aligned} \mathbb{E} \sigma_d \left(\sum_{i \in H} \epsilon_i m_i \right)^2 &\geq \mathbb{E} \sum_{k \in H} \left| \sum_{i \in H} \epsilon_i \langle F_k, m_i \rangle \right|^2 \\ &= \sum_{k \in H} \sum_{i \in H} |\langle F_k, m_i \rangle|^2 \geq |H| \delta^2. \end{aligned}$$

The equality uses cancellation of cross terms under the independent signs; no cross-pairing is assumed to vanish. By (6.5),

$$\max_{\epsilon_i \in \{-1, 1\}} \left\| \sum_{i \in H} \epsilon_i m_i \right\|_Z^2 \geq \mathbb{E} \left\| \sum_{i \in H} \epsilon_i m_i \right\|_Z^2 \geq \alpha_d |H| \delta^2.$$

Taking square roots proves (7.5). $\square$

It remains to make the detectors from Lemma 7.1 compatible without losing their nonzero pairings. We will use two support configurations: disjoint supports, or later supports contained in cylinders where earlier functions are constant.

7.3 Support conditions for compatibility

Localizing detecting Lipschitz functions and combining them with a common Lipschitz bound is also central to Kalton's arguments for metric free spaces [9, Lemma 4.5 and proof of Theorem 4.6]. Here the supports are finite-coordinate cylinders and tail regions, and their compatibility must be retained within the selected test classes defining Z .

Lemma 7.3 (Compatible supports). *Let B be a convex subset of a normed space. Let u_i, b_i be continuous real functions on B , with b_i Lipschitz, and put $F_i = u_i b_i$. Suppose $\text{Lip}(F_i) \leq L$ for every i .*

1. *If the closed relative supports $\text{supp}_B F_i$ are pairwise disjoint, every finite signed sum of the F_i has Lipschitz constant at most L .*
2. *Suppose that for each i there is a relatively open set $C_i \subset B$ and a scalar e_i such that $u_i = e_i$ on C_i and $\text{supp}_B u_k \subset C_i$ for every $k > i$. If $\sum_i |e_i| \text{Lip}(b_i) < \infty$, every finite signed sum of the F_i has Lipschitz constant at most*

$$L + \sum_i |e_i| \text{Lip}(b_i).$$

If $b_i = 1$ for every i , the summability condition is automatic.

Proof. Fix a finite set of indices and signs. In the first case, at a given point at most one of the corresponding closed supports contains that point. All other summands vanish on a relative neighborhood, so the sum is locally either zero or a single signed F_i , with bound L .

In the second case, choose at a given point the largest selected index j whose u_j -support contains the point. If there is no such index, every summand vanishes on a common neighborhood, because there are only finitely many closed supports. Otherwise all selected later functions vanish on a neighborhood. The point lies in C_i for each selected $i < j$, so on a smaller neighborhood all those u_i equal e_i . On that neighborhood the sum is

$$\epsilon_j F_j + \sum_{\substack{i < j \\ i \text{ selected}}} \epsilon_i e_i b_i,$$

and has the asserted local Lipschitz bound.

To pass from the uniform local bound to a global bound, take any two points of B . The segment between them is contained in B . Its compact parameter interval has a finite cover by the relative neighborhoods just obtained and hence a Lebesgue number. Subdivide the segment finely enough that each subsegment lies in one such neighborhood, apply the local estimate to each subsegment, and add. The lengths of the collinear subsegments sum to the distance of the endpoints. This proves both global assertions. $\square$

Lemma 7.4 (Tail separation). *Suppose F_i belong to one listed class and have representations with bounded band indices. Suppose $m_i \in Z$ and*

$$|\langle F_i, m_i \rangle| \geq \delta > 0, \quad \text{supp}_B F_i \subset \{\tau_{M_i} \geq c_*\}, \quad M_i \in \mathbb{N}, \quad M_i \rightarrow \infty, \quad c_* > 0. \quad (7.6)$$

Then a subsequence has modified tests in one class $\mathcal{V}_H$, with detection at least $\delta/2$, whose relative supports are pairwise disjoint.

Proof. Choose a dyadic $c = 2^{-a}$ with $2c < c_*$. If the original class is $\mathcal{T}_h$, view a test as a cylindrical factor with no tail factors; if it is $\mathcal{V}_h$, keep its chosen product representation. A single integer H , larger than h , the bounded band indices, and a , can be chosen so that both F_i and $F_i b_{c,N}$ belong to $\mathcal{V}_H$ for all i, N . There is room for one extra tail factor.

For each fixed i , $b_{c,N}(s) \rightarrow 1$ at every $s \in B$, since $\tau_N(s) \rightarrow 0$. Lemma 6.2, used in this one class, therefore gives $\langle F_i b_{c,N}, m_i \rangle \rightarrow \langle F_i, m_i \rangle$. Choose N large enough that the difference has modulus less than $\delta/2$; every sufficiently large N then works. This choice is made for the fixed vector m_i .

Select indices recursively. After choosing an index i and such an N_i , take the next original index so far out that its M_k exceeds all previously selected N_i , which is possible because $M_k \rightarrow \infty$. Then choose its own N_k as above. For selected $k > i$, the new i -th support requires $\tau_{N_i} \leq 2c$ by (5.10). The new k -th support requires $\tau_{M_k} \geq c_*$ by (7.6), hence $\tau_{N_i} \geq \tau_{M_k} \geq c_* > 2c$. The supports are disjoint, and the detection bound follows from the choice of N_i . $\square$

7.4 Selecting compatible detectors

The next proposition isolates the geometric part of the argument. It uses only weak nullity of the vectors, one fixed test class, and a uniform lower bound on the pairings. The bounded signed sums will enter again only when we apply Lemma 7.2.

Proposition 7.5 (Selection of compatible detectors). *Let $(m_i) \subset Z$ be weakly null, meaning $u(m_i) \rightarrow 0$ for every $u \in Z^*$. Suppose F_i belong to one listed class $\mathcal{L}$ and*

$$|\langle F_i, m_i \rangle| \geq \delta > 0 \quad (i \geq 1). \quad (7.7)$$

There are a subsequence (m_{i_k}) , a listed class $\mathcal{L}_1$, tests $\tilde{F}_k \in \mathcal{L}_1$, $\delta_1 > 0$, and an integer $C_1 \geq 1$ such that

$$|\langle \tilde{F}_k, m_{i_k} \rangle| \geq \delta_1, \quad \text{Lip} \left(\sum_{k \in H} \epsilon_k \tilde{F}_k \right) \leq C_1$$

for every k , every finite H , and all signs ϵ_k .

Proof. We pass to subsequences and relabel their indices throughout the proof. Every fixed test pairs to zero in the limit, by weak nullity and Lemma 6.2.

Fix a concrete representation of each detector F_i . If its class is $\mathcal{T}_h$, take $F_i = u_i$ and no tail factor. If its class is $\mathcal{V}_h$, write $F_i = u_i b_i$, where u_i is its chosen cylindrical factor and b_i is its product of tail factors. Let $\gamma_i = (\lambda_i, j_i, \xi_i)$ be the chosen label. After subsequences, either the labels are pairwise distinct or one label is repeated throughout. In the distinct case the bands have a bounded subsequence or a subsequence tending to infinity. We handle these three possibilities separately.

Distinct labels with bounded bands. Every block Γ_λ is finite, so pairwise distinct labels force λ_i to leave every finite set. Since $\{\lambda : J_\lambda \leq J\}$ is finite for every J , we have $J_{\lambda_i} \rightarrow \infty$. Pass to a further subsequence with one fixed band j . Eventually $j < J_{\lambda_i}$, and (5.8) gives

$$r_{\lambda_i} \geq c_* := \ell_j/2 > 0 \quad \text{on } \text{supp}_B F_i. \quad (7.8)$$

The coordinates used by these radii escape every fixed initial segment. Indeed, each coordinate occurs in at most two radii, so a finite set of coordinates occurs in only finitely many levels. After discarding finitely many terms, choose integers $M_i \rightarrow \infty$ so that every coordinate used by r_{λ_i} has index $> M_i$. Then $r_{\lambda_i}(s) \leq \tau_{M_i}(s)$. Condition (7.8) is the hypothesis of Lemma 7.4, which gives detectors in one enlarged class with pairwise disjoint supports. Lemma 7.3(1) gives a common Lipschitz bound for all their finite signed sums.

Distinct labels with bands tending to infinity. Now take $j_i \rightarrow \infty$. The fixed class must be $\mathcal{T}_h$, because a $\mathcal{V}_h$ representation has $j \leq h$. By diagonal extraction from the compact intervals $[-1, 1]$, the finite tags $\xi_i \in [-1, 1]^{j_i}$ converge coordinatewise to some $v_* \in [-1, 1]^{\mathbb{N}}$. This point need not be in c_0 . Only its finite coordinate restrictions will be used.

Write $a_i = 2d_{j_i} + \ell_{j_i}/2$, which tends to zero. For every fixed nonempty finite coordinate set I , once $j_i \geq \max I$, (5.8) gives

$$\sup_{s \in \text{supp}_B F_i} \|s|_I - v_*|_I\|_\infty \leq a_i + \|\xi_i|_I - v_*|_I\|_\infty \rightarrow 0. \quad (7.9)$$

For each fixed i , consider arbitrarily small positive dyadic scales θ , below the last existing scale if there is one and with θ less than the remaining strict budget in (5.6). Choose a grid center $q_i(\theta)$ within $\theta/8$ of $v_*|_{I_{\gamma_i}}$. The test

$$F_{i,\theta} = G_{q_i(\theta),\theta} F_i$$

belongs to $\mathcal{T}_{h+1}$, as does F_i , and

$$\|F_{i,\theta} - F_i\|_\infty \leq \theta \text{Lip}(F_i) \leq \theta 4^h L_0.$$

Thus $F_{i,\theta} \rightarrow F_i$ pointwise as $\theta \downarrow 0$, even though the centers move. For this fixed i , Lemma 6.2 and (6.8) allow a choice θ_i with

$$|\langle F_{i,\theta_i} - F_i, m_i \rangle| < \delta/2.$$

This is a separate choice after fixing m_i , not a uniform limit over the sequence of vectors. Write $\tilde{F}_i = F_{i,\theta_i}$. It detects m_i with bound $\delta/2$.

The function $\tilde{F}_i$ is constant on the relatively open cylinder

$$C_i = \{s \in B : \|s|_{I_{\gamma_i}} - v_*|_{I_{\gamma_i}}\|_\infty < 3\theta_i/8\}, \quad (7.10)$$

because the distance from its grid center is then less than $\theta_i/2$. Its enlarged representation still satisfies the strict budget, so (5.8) also applies to $\tilde{F}_i$. Therefore (7.9) holds with $\tilde{F}_i$ in place of F_i . For each fixed earlier i , all sufficiently late supports lie in C_i . A recursive subsequence makes this true for every later selected index simultaneously. Now Lemma 7.3(2), with $u_i = \tilde{F}_i$ and $b_i = 1$, gives a common Lipschitz bound for finite signed sums.

One repeated label. It remains to treat a fixed label $\gamma = (\lambda, j, \xi)$. Here all cylindrical factors depend on the same finite set $I = I_\gamma$. We first expand the tests to isolate a varying term that still detects the vectors. An escaping tail index then gives disjoint supports. Otherwise, a first small remainder R gives shrinking cylindrical supports; a final local modification produces nested plateaux with summably small values.

Extract a subsequence on which the number $m \leq h$ and the types G, R of cylindrical operations are fixed, as are the number $l \leq h$ and the types of tail factors. For each operation position its dyadic scale has a subsequence that is either constant or tends to zero. In the constant case its center can also be made constant, because the permitted grid in the fixed

cube is finite. At each tail position the scale c can be made constant, and its integer index N is either constant or tends to infinity. Make all these finitely many extractions together.

Expansion and detection. Call an ingredient fixed when all of its parameters are now constant. The parameters in each concrete representation are frozen before the following algebraic expansion. Replace every nonfixed $G_{q,\varepsilon}$ by $\text{Id} + R_{q,\varepsilon}$, and every nonfixed factor $b_{c,N}$ by $1 + (b_{c,N} - 1)$. The operations are linear in their input functions, so these identities distribute through later operations with no parameters recomputed. The expansion is only of the cylindrical operators and of the tail product separately; it never applies G to a tail-dependent full test.

There are at most $K_0 = 2^{m+l} \leq 2^{2h}$ resulting term positions. Each cylindrical term retains a chronological subsequence of the original operations, possibly replacing G by R at the same scale. Its scales remain nonincreasing and its total scale does not increase. Omitting a tail factor or replacing b by $b - 1$ also preserves the permitted type and scale. Because j is fixed, all resulting full terms belong to one class $\mathcal{V}_H$ for some fixed $H \geq \max\{h, j\}$, even if the original class was $\mathcal{T}_h$. Write the exact expansion as

$$F_i = \sum_{a=1}^K F_i^a, \quad K \leq K_0. \quad (7.11)$$

A term position containing only fixed ingredients is a fixed concrete test F^a . Its pairing with m_i tends to zero by weak nullity; it is not asserted to vanish identically. There are only finitely many such positions, so their total pairing tends to zero. In view of (7.7), for all sufficiently large i the sum of the absolute pairings of the varying terms in (7.11) is at least $\delta/2$. One of the at most K_0 positions has pairing at least $\delta/(2K_0)$; a subsequence fixes that position. Relabel its tests as

$$F_i = u_i b_i, \quad |\langle F_i, m_i \rangle| \geq \delta_0 > 0. \quad (7.12)$$

If b_i contains a nonfixed factor $b_{c,N_i} - 1$ with $N_i \rightarrow \infty$, its support is contained in $\{\tau_{N_i} \geq c\}$ by (5.10). The band is fixed, so Lemma 7.4 applies. We may therefore assume no such factor remains. Every retained nonfixed tail factor would have this form after expansion. The remaining tail product is therefore fixed, and the variation in (7.12) must occur in the cylindrical factor. Every retained nonfixed cylindrical operation is an R : a nonfixed G was replaced by an omitted identity or by R .

A small remainder. Choose the first retained R whose scale $\varepsilon_i \rightarrow 0$, and call its center q_i . All retained operations before it are fixed, so they produce one fixed prefix function p from f_γ . Set

$$L_{\text{pre}} = \text{Lip}(p) \leq 4^h L_0.$$

This is the Lipschitz bound of the prefix, not the base bound L_0 ; earlier operations may have enlarged it. If $L_{\text{pre}} = 0$, this R and the resulting test vanish, contradicting (7.12). Lemma 5.1 gives

$$\|R_{q_i, \varepsilon_i} p\|_\infty \leq L_{\text{pre}} \varepsilon_i, \quad K_I(R_{q_i, \varepsilon_i} p) \subset \overline{B}_\infty(q_i, \varepsilon_i).$$

Every retained later scale is at most ε_i , by the nonincreasing order in the original representation. A later G does not increase the supremum norm, a later R increases it by at most a factor two, and either enlarges projected support by at most its scale. Iterating the two parts of Lemma 5.1 over at most h later operations yields

$$\|u_i\|_\infty \leq 2^h L_{\text{pre}} \varepsilon_i, \quad \{s|_I : s \in \text{supp}_B u_i\} \subset K_I(u_i) \subset \overline{B}_\infty(q_i, (h+1)\varepsilon_i). \quad (7.13)$$

The first inclusion follows by continuity of coordinate restriction and the definition of relative support. This is where the order of scales is essential: a fixed total budget alone would not make the sum of the later scales tend to zero.

Pass to a subsequence with $q_i \rightarrow v$ in the compact cube $[-1, 1]^I$. We now make each cylindrical factor constant on a small cylinder about v , while preserving (7.12). For a fixed i , choose arbitrarily small positive dyadic θ that are below the last retained scale, below the remaining strict budget, and at most ε_i . Let $q_i^+(\theta)$ be a permitted grid point within $\theta/8$ of v , and set

$$\tilde{F}_{i,\theta} = (G_{q_i^+(\theta),\theta} u_i) b_i.$$

All these full tests, and $F_i = u_i b_i$, belong to one fixed enlarged class $\mathcal{V}_{H'}$, with H' large enough for one more cylindrical operation. The tail product has supremum norm at most one, so

$$\|\tilde{F}_{i,\theta} - F_i\|_\infty \leq \theta \text{Lip}(u_i).$$

For the fixed vector m_i , pointwise convergence and (6.8) give an allowed choice θ_i such that

$$|\langle \tilde{F}_{i,\theta_i} - F_i, m_i \rangle| < \delta_0/2. \quad (7.14)$$

Again the scale is chosen separately for each m_i ; no uniform dual convergence on the moving sequence is used.

Write $\tilde{u}_i = G_{q_i^+(\theta_i),\theta_i} u_i$ and $\tilde{F}_i = \tilde{u}_i b_i$. It detects with bound $\delta_0/2$. The support enlargement in Lemma 5.1, (7.13), and $\theta_i \leq \varepsilon_i$ give

$$\{s|_I : s \in \text{supp}_B \tilde{u}_i\} \subset \overline{B}_\infty(q_i, (h+2)\varepsilon_i). \quad (7.15)$$

This conclusion does not require the new grid center to be close to q_i : a new nonzero value can be created only near an old nonzero value, which is exactly the support enlargement proved earlier. As $q_i \rightarrow v$ and $\varepsilon_i \rightarrow 0$, these cylindrical supports concentrate at v .

On the relatively open cylinder

$$C_i = \{s \in B : \|s|_I - v\|_\infty < 3\theta_i/8\},$$

the function $\tilde{u}_i$ is the constant $e_i = u_i(q_i^+(\theta_i))$. Equation (7.13) gives

$$|e_i| \leq 2^h L_{\text{pre}} \varepsilon_i \rightarrow 0. \quad (7.16)$$

Nested plateaux. Choose a recursive subsequence on which $\sum_i |e_i| < \infty$ and $\text{supp}_B \tilde{u}_k \subset C_i$ whenever $k > i$. To see that both requirements can be imposed, after selecting finitely many indices use (7.15) to discard a finite initial segment so that all remaining supports lie in their cylinders; then choose the next index with, for example, $|e_i| \leq 2^{-i}$, using (7.16). Continue.

The remaining b_i are products of at most h tail factors of supremum norm at most one and Lipschitz constant at most 2^h . Thus $\text{Lip}(b_i) \leq h2^h$ uniformly. The full $\tilde{F}_i$ have a common Lipschitz bound because they are in $\mathcal{V}_{H'}$, and

$$\sum_i |e_i| \text{Lip}(b_i) < \infty.$$

Lemma 7.3(2), applied to the cylindrical supports in (7.15), gives the required common Lipschitz bound for all finite signed sums. The nesting is of $\text{supp}_B \tilde{u}_i$, not merely the possibly smaller supports of $\tilde{F}_i$.

In each case the modified tests remain in one listed class, retain a positive common lower bound on their pairings, and have a common Lipschitz bound for finite signed sums. Choose an integer C_1 no smaller than this last bound. These are the required conclusions. $\square$

7.5 Excluding a linear copy of c_0

Proposition 7.6. *The Banach space Z contains no linearly isomorphic copy of c_0 .*

Proof. Suppose $T : c_0 \rightarrow Z$ is an isomorphic embedding and put $m_i = Te_i$. The lower bound for T and the identity $\|\sum_{i \in H} \epsilon_i e_i\|_\infty = 1$ for nonempty finite H give (7.1) for some $r, M > 0$. Lemma 7.1 supplies a weakly null subsequence and detectors in one class. Proposition 7.5 then supplies compatible detectors with a common nonzero pairing. For their class $\mathcal{L}_1$, bound C_1 , and pairing constant δ_1 , set $d_1 = (\mathcal{L}_1, C_1, 2)$. Lemma 7.2 gives

$$M^2 \geq \alpha_{d_1} |H| \delta_1^2$$

for every finite set of selected indices H , a contradiction. Thus no such embedding T exists. $\square$

8 Assembly and same-space consequences

The construction now supplies both inputs needed after the absorption criterion: a lifted correction and the absence of linear c_0 .

Proof of Theorem 1.1. Lemma 6.1 gives a separable real Banach space Z . Proposition 7.6 excludes a linearly isomorphic copy of ordinary c_0 in Z . Propositions 4.2 and 4.3 give a bi-Lipschitz bijection

$$g : E \oplus_\infty U \longrightarrow U$$

with $E = c_0(\mathbb{N})$ and lower constant $c_g = 1/5 - 24\eta > 0$. Lemmas 6.3 and 6.4 give a bounded linear $Q : Z \rightarrow U$ and a Lipschitz $K : E \oplus_\infty U \rightarrow Z$ with $\text{Lip}(K) \leq 3$ and $QK = g - P$. Applying Lemma 2.1 yields the onto bi-Lipschitz map

$$F : Z \oplus_\infty E \rightarrow Z, \quad F(b, t) = b + K(t, Qb),$$

with the explicit inverse (2.3). Its inverse is the bi-Lipschitz bijection in the opposite direction $Z \rightarrow Z \oplus_\infty c_0$. This proves the theorem. $\square$

Corollary 8.1. *For this same separable real Banach space Z :*

1. $Z \oplus_\infty c_0$ and Z are bi-Lipschitz equivalent but not linearly isomorphic.
2. c_0 embeds bi-Lipschitzly into Z , although Z contains no linear copy of c_0 .
3. Every separable metric space embeds bi-Lipschitzly into Z .

Proof. The direct summand $\{0\} \oplus c_0$ is a linear isometric copy of c_0 . A bounded linear isomorphism from the direct sum onto Z would carry it to a closed linear copy in Z , contradicting Proposition 7.6. The restriction $t \mapsto F(0, t)$ is a bi-Lipschitz embedding, proving the second statement. Aharoni's theorem [1] embeds every separable metric space bi-Lipschitzly into c_0 ; composition with this restriction gives the third statement. $\square$

References

- [1] Israel Aharoni. Every separable metric space is Lipschitz equivalent to a subset of c_0^+ . *Israel Journal of Mathematics*, 19:284–291, 1974.
- [2] Israel Aharoni and Joram Lindenstrauss. Uniform equivalence between Banach spaces. *Bulletin of the American Mathematical Society*, 84(2):281–283, 1978.

- [3] Israel Aharoni and Joram Lindenstrauss. An extension of a result of Ribe. *Israel Journal of Mathematics*, 52(1–2):59–64, 1985.
- [4] Richard F. Arens and James Eells, Jr. On embedding uniform and topological spaces. *Pacific Journal of Mathematics*, 6(3):397–403, 1956.
- [5] Gilles Godefroy and Nigel J. Kalton. Lipschitz-free Banach spaces. *Studia Mathematica*, 159(1):121–141, 2003.
- [6] Gilles Godefroy, Nigel J. Kalton, and Gilles Lancien. Subspaces of $c_0(\mathbb{N})$ and Lipschitz isomorphisms. *Geometric and Functional Analysis*, 10(4):798–820, 2000.
- [7] Petr Hájek, Michal Johannis, and Thomas Schlumprecht. Remarks on the point character of Banach spaces and non-linear embeddings into $c_0(\Gamma)$. *Pure and Applied Functional Analysis*, 10(4):811–824, 2025.
- [8] William B. Johnson, Joram Lindenstrauss, and Gideon Schechtman. Banach spaces determined by their uniform structures. *Geometric and Functional Analysis*, 6(3):430–470, 1996. Theorem locators refer to <https://arxiv.org/abs/math/9701203v1>.
- [9] Nigel J. Kalton. Spaces of Lipschitz and Hölder functions and their applications. *Collectanea Mathematica*, 55(2):171–217, 2004.
- [10] Nigel J. Kalton. The nonlinear geometry of Banach spaces. *Revista Matemática Complutense*, 21(1):7–60, 2008.
- [11] OpenAI. Lipschitz Equivalent Separable Banach Spaces Need Not Be Linearly Isomorphic. OpenAI Math Release preprint [OAI:Lipschitz-Equivalent-Separable-Banach-Spaces-Need-Not-Be-Linearly-Isomorphic-September-24-2026](#), 2026.
- [12] Martin Ribe. On uniformly homeomorphic normed spaces. *Arkiv för Matematik*, 14:237–244, 1976.
- [13] Martin Ribe. Existence of separable uniformly homeomorphic nonisomorphic Banach spaces. *Israel Journal of Mathematics*, 48(2–3):139–147, 1984.
- [14] Bünyamin Sarı. A coarse-Lipschitz embedding of c_0 into a separable dual Banach space, 2026. arXiv:2608.04117v2, 26 August 2026.